

Sneha Mishra

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LinkedIn | GitHub | LeetCode

Summary

Software Engineering undergraduate with strong foundations in Data Structures & Algorithms, Object-Oriented Programming, and Backend Development. Experienced in building scalable RESTful APIs and full-stack applications using Node.js, Express.js, MongoDB, MySQL, and React.js. Passionate about designing efficient backend systems, optimizing database performance, and building reliable software for large-scale applications.

Experience

Tata Steel

May 2026 – Jun 2026

Research Intern

- Optimized heat flux parameters in Thin Slab Casting and Rolling under the Head of TSCR Department, reducing crack breakout incidents by approximately 12% across multiple production runs.
- Utilized Python for data analysis and heat transfer evaluation to identify process parameters affecting casting quality.
- Systematically debugged and resolved complex data-analysis issues by identifying incorrect input data and handling exceptions to optimize the analysis workflow.

Education

Indian Institute of Engineering Science and Technology, Shibpur

Aug 2023 – Present

Bachelor of Technology

CGPA: 8.26/10

Indian Institute of Technology Madras

May 2024 – May 2025

Foundation Level, Data Science

Positions of Responsibility

Braking Lead – BAJA SAE Team, IEST Shibpur

May 2025 – Present

- Led the design and analysis of the braking system for the BAJA SAE vehicle, ensuring compliance with safety and performance standards.
- Collaborated with multidisciplinary teams to design, test, troubleshoot, and optimize the braking subsystem throughout the development cycle.

Technical Skills

Programming: Java, C++, Python, JavaScript, TypeScript

Backend: Node.js, Express.js, RESTful APIs, JWT Authentication, MVC Architecture

Databases: MySQL, MongoDB

Frontend: React.js, Next.js, HTML, CSS

Core CS: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS

Libraries: NumPy, Pandas, Scikit-learn

Tools: Git, GitHub, Postman

Projects

Hostel Complaint & Application Management System | *Node.js, Express.js, MongoDB*

- Developed a scalable full-stack hostel management system using Node.js, Express.js, and MongoDB.
- Designed and implemented RESTful APIs for complaint management, application processing, and administrative operations.
- Implemented JWT-based authentication and Role-Based Access Control (RBAC) for secure access management.
- Designed MongoDB collections and optimized database queries to improve backend performance.
- Followed MVC architecture to build maintainable and scalable backend services.
- Applied strong understanding of backend service development, RESTful APIs, and MySQL database management, with a high aptitude for rapidly transitioning to PHP-based server-side development.
- Optimized search performance by indexing critical data in Elasticsearch, reducing query latency for large-scale administrative data retrieval.

Credit Card Fraud Detection Using Machine Learning | *Python, Scikit-learn, Pandas, NumPy*

- Built a machine learning model using Logistic Regression to detect fraudulent credit card transactions.
- Performed data preprocessing, feature engineering, and handled class imbalance through undersampling.
- Achieved 97.48% training accuracy and 98.32% testing accuracy.
- Built an end-to-end prediction pipeline for automated fraud detection and transaction classification.